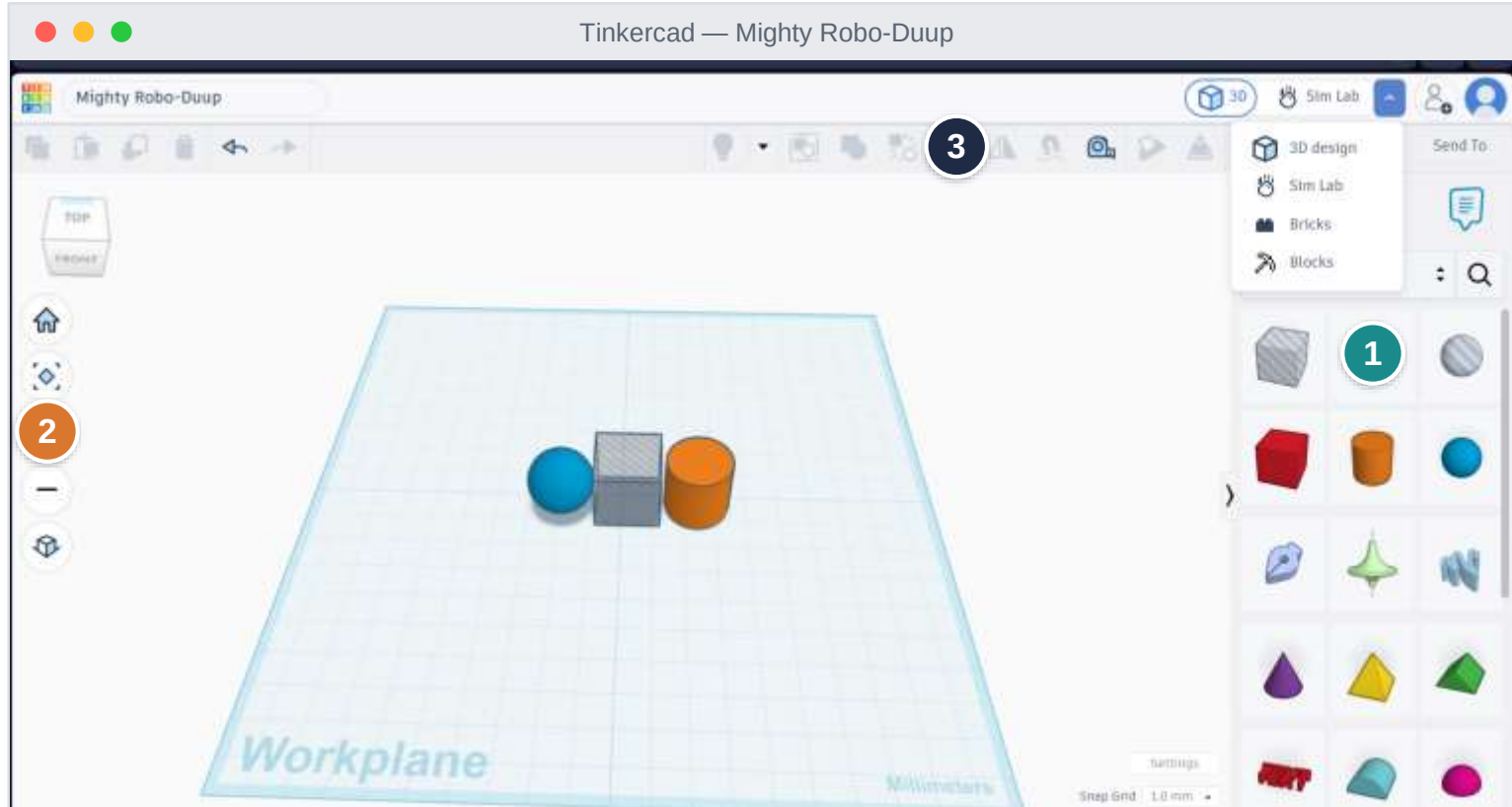


HCI Focus — Tinkercad Interaction Elements



1 Shape Library

Drag-and-drop 3D primitives onto the workplane, lowering the skill barrier for beginners.

2 Navigation Controls

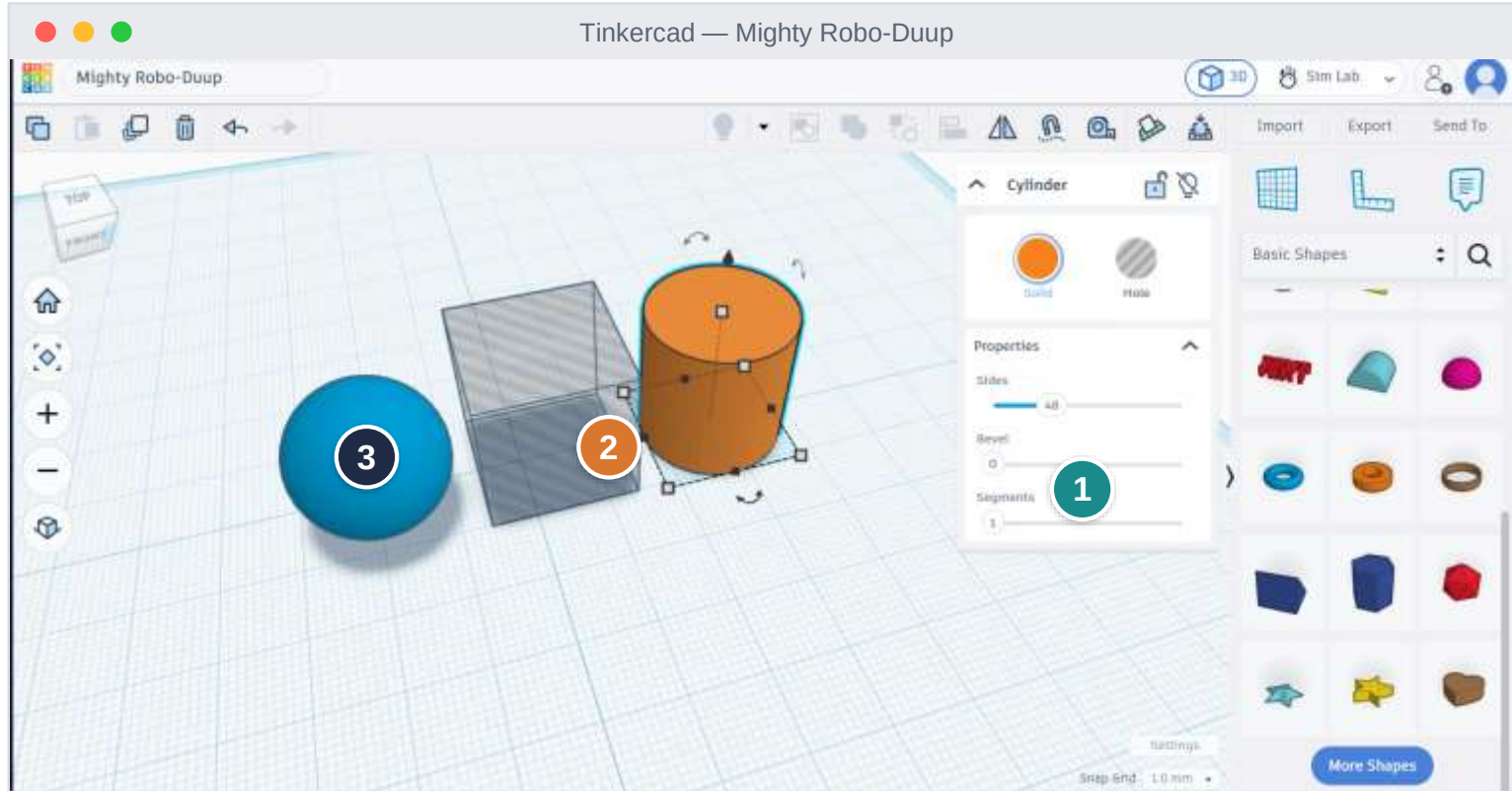
Home / zoom / orbit icons give direct camera manipulation to inspect the model from any angle.

3 Top Toolbar

Icon-based quick actions (undo, align, mirror) give instant visual feedback on available edits.

Reflection — HCI Change: I'd add a hover tooltip on each shape icon in the sidebar (e.g. "Cylinder – click to add"), since new users may not know what an icon does until after they've already dragged it onto the workplane.

CG Focus — Tinkercad Rendering Elements



1 Mesh Geometry

The Segments/Sides panel shows the cylinder is built from a defined polygon count — how curved surfaces are approximated with flat faces.

2 Perspective Camera View

The workplane converges toward a vanishing point, giving depth cues that make the flat screen read as 3D space.

3 Shading & Light Source

Gradient shading across the sphere/cylinder and a darker cube face reveal a single light source, creating the illusion of volume.

Reflection — CG & Interaction: Tinkercad's simple, flat-shaded CG actually helps interaction — objects stay easy to read and never distract from precise placement. Photorealistic lighting could instead soften edges and make it harder to judge exact alignment.